LIHEAP and Food Assistance

Correlation Statistical Report

San Diego County zip codes, February to October 2023 | San Diego Data Science Alliance | October 2026

Executive Summary: Key Findings

- **Strong, consistent link.** LIHEAP enrollment tracks CalFresh dollars closely (Spearman ρ up to 0.70) and food bank pounds moderately (ρ ≈ 0.46).
- **Where, not when.** No time lag is clearly best. LIHEAP marks which zip codes need food assistance, not when need will rise.
- **Robust to policy.** Correlations held through every policy change and when any single month was dropped.
- **Not just zip size, but poverty matters.** Per household, correlations fall by 0.10 to 0.20 but stay moderate. The food bank link holds only in high-poverty zip codes.

Data and Methods

The analysis merges zip-by-month LIHEAP enrollment counts, CalFresh benefit dollars and food bank pounds (852 zip-months), with LIHEAP also shifted back one to three months (lags 1 to 3). Food bank months with zero pounds were treated as missing.

We report Pearson r and Spearman ρ with 95% confidence intervals (Fisher z; Bonett-Wright for Spearman), re-estimated on a common sample complete at all lags (268 CalFresh, 262 food bank zip-months). Within-zip correlations, computed after subtracting each zip's average, isolate month-to-month co-movement. Groups were compared with Fisher z-tests, and a leave-one-month-out analysis tested sensitivity.

Strongest Correlations Identified

All correlations were positive and significant at $p < 0.001$, and CalFresh was more strongly related than food bank distribution at every lag.

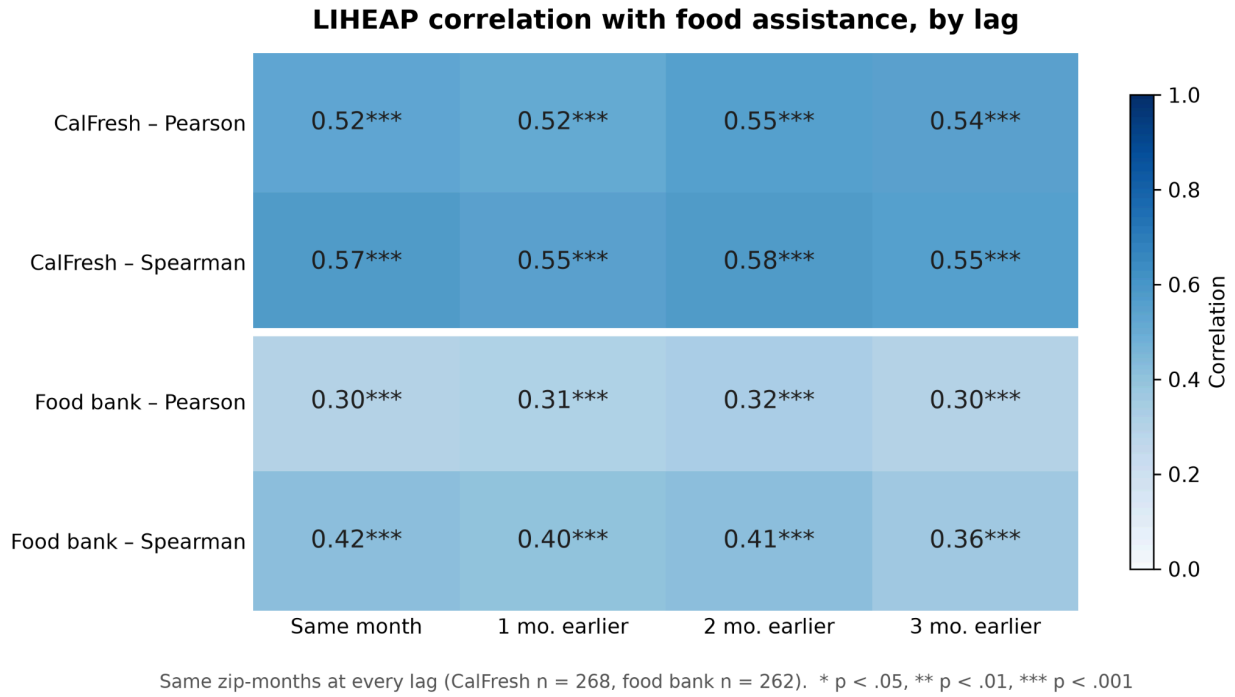
Outcome	Lag	n	Pearson r [95% CI]	Spearman ρ [95% CI]
CalFresh	Same month	453	0.57 [0.50, 0.63]	0.64 [0.58, 0.70]
CalFresh	1 month earlier	577	0.57 [0.51, 0.62]	0.68 [0.63, 0.73]
CalFresh	2 months earlier	578	0.55 [0.49, 0.60]	0.68 [0.63, 0.73]
CalFresh	3 months earlier	577	0.59 [0.54, 0.64]	**0.70 [0.65, 0.74]**
Food bank	Same month	419	0.35 [0.26, 0.43]	**0.47 [0.39, 0.55]**
Food bank	1 month earlier	502	0.34 [0.26, 0.41]	0.46 [0.38, 0.53]

Food bank	2 months earlier	504	0.33 [0.25, 0.41]	0.46 [0.38, 0.53]
Food bank	3 months earlier	504	0.35 [0.27, 0.42]	0.46 [0.38, 0.53]

The strongest single correlation was LIHEAP three months earlier with CalFresh dollars (ρ = 0.70, n = 577). For food bank pounds the strongest was same-month enrollment (ρ = 0.47). Spearman exceeded Pearson at every lag because a few high-volume zip codes sit off a straight line.

Time Lag with Highest Predictive Value

No lag has clearly higher predictive value. Confidence intervals overlap, and on the common sample every lag falls within 0.06 of the others (Figure 1).



Same zip-months at every lag (CalFresh n = 268, food bank n = 262). * p < .05, ** p < .01, *** p < .001

Figure 1. *Correlations are nearly flat across lags on the same zip-months, so timing adds little predictive value.*

Within-zip r	Lag 0	Lag 1	Lag 2	Lag 3
CalFresh (n = 268)	**0.53** (p < .001)	0.44 (p < .001)	−0.12 (p = .055)	−0.27 (p < .001)
Food bank (n = 262)	−0.03 (p = .59)	0.07 (p = .30)	0.05 (p = .45)	0.15 (p = .013)

Within zips, LIHEAP moves with CalFresh dollars in the same month (r = 0.53) and one month later (r = 0.44), so **the one-month lag is the most useful leading indicator for CalFresh**. Food bank pounds show no reliable month-to-month link (lag-3 p = 0.013 fails Bonferroni). Lag 1 is used for the policy and sensitivity analyses.

Geographic and Demographic Segments

Within-zip Spearman correlations (same-month LIHEAP) were computed for 71 of 121 zip codes with at least five months of data, and they varied widely.

Strongest	Avg ρ	Weakest or negative	Avg ρ
92154 San Diego	**0.90**	91911 Chula Vista	**−0.71**
92082 Valley Center	**0.87**	91915 Chula Vista	**−0.71**
92122 San Diego (CalFresh only)	**0.87**	91914 Chula Vista (CalFresh only)	**−0.65**
91910 Chula Vista	0.72	92064 Poway	−0.40
91977 Spring Valley	0.71	92108 San Diego	−0.31
92126 San Diego	0.69	91932 Imperial Beach	−0.30
91950 National City	0.67	92081 Vista	−0.28

The strongest relationships appear in South Bay and inland zips with large caseloads, while Chula Vista spans both ends of the ranking. With five or six months per zip and about 140 tests, about seven would reach $p < 0.05$ by chance; only 92154 was significant for both programs. Treat the ranking as a list for closer review.

Demographic tiers. Zip codes were split into thirds on six ACS measures (poverty, unemployment, rent, share aged 65+, single-parent households, household size), and high and low thirds were compared with Bonferroni adjustment for 12 tests ($p < 0.0042$). Thirty-two mostly PO-box zips had no census match.

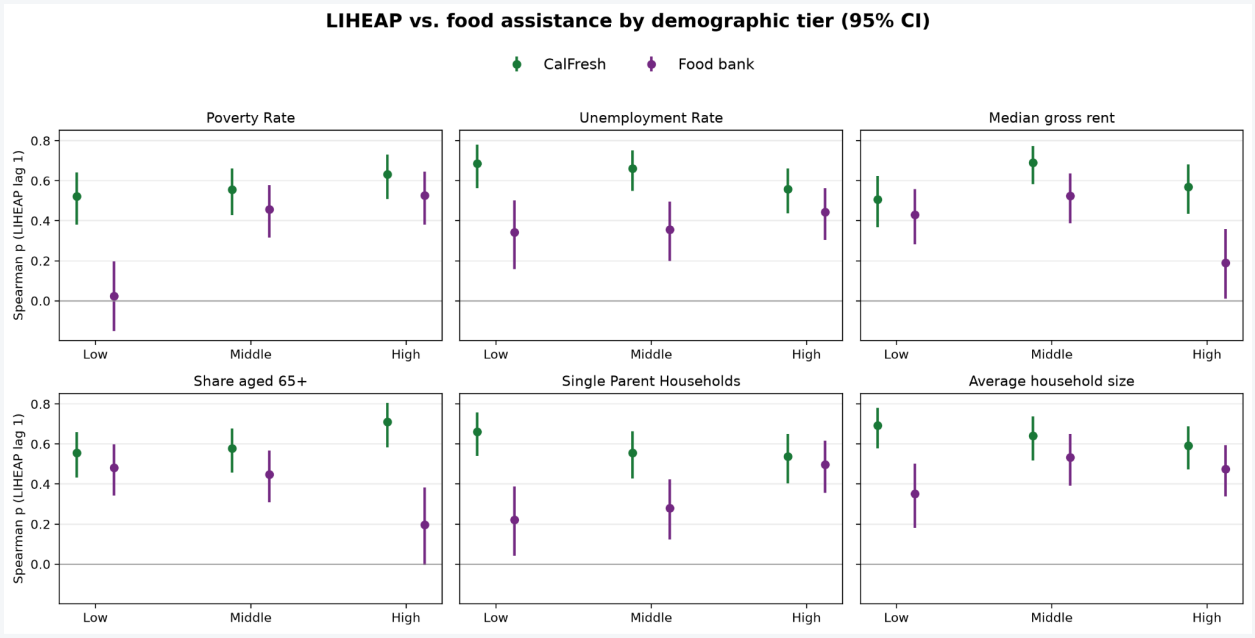


Figure 2. CalFresh correlations (green) are similar in every tier; food bank correlations (purple) collapse in

low-poverty, high-rent and older zip codes.

Only poverty survives correction, and only for food bank. Food bank pounds track LIHEAP in high-poverty zips (above 11.2%: ρ = 0.53, n = 133) but not low-poverty ones (below 8.0%: ρ = 0.02, n = 128; p < 0.001). Single-parent share, age 65+ and rent point the same way (p = 0.012 to 0.036) but do not survive correction. Pantries likely cluster in higher-need areas, so in low-poverty zips pounds reflect site locations more than demand. CalFresh correlations were similar in every group (ρ = 0.51 to 0.71), and demographics explain little of the zip ranking (strongest: rent, ρ = −0.30).

Impact of Policy Changes

Using LIHEAP one month earlier, correlations were recomputed for policy-stable and policy-change months, and for months with and without pandemic emergency provisions.

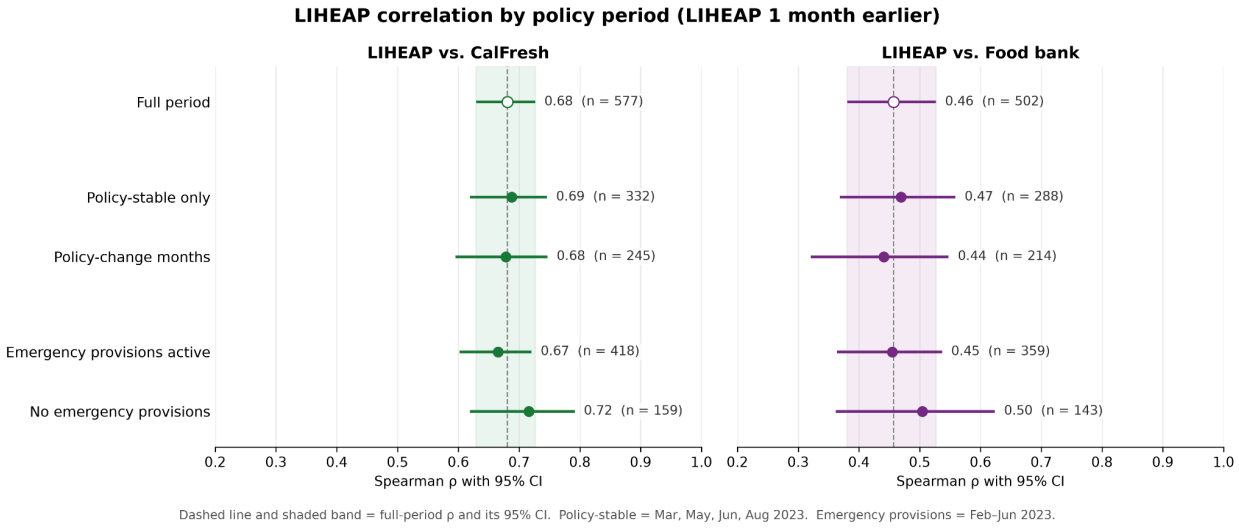


Figure 3. *Every policy period falls inside the full-period confidence band (shaded) for both programs.*

None of the regime differences was statistically significant. Policy-change and stable months differed by −0.01 for CalFresh (p = 0.84) and −0.03 for food bank (p = 0.71), and correlations were slightly higher after emergency provisions ended (p = 0.32 and 0.53). Policy changed how much assistance zip codes received more than which received the most. At 80% power these tests detect only differences of about 0.14 (CalFresh) and 0.22 (food bank), so they rule out large shifts, not all effects.

Sensitivity Analysis

No single month drives the results. Dropping each month in turn moved CalFresh ρ between 0.666 and 0.705 (full period 0.681) and food bank ρ between 0.447 and 0.473 (full period 0.457), with policy-change and stable months producing similar shifts.

Statistical Limitations and Caveats

Correlation is not causation. These results show that LIHEAP enrollment and food assistance occur together, not that one program drives the other.

Zip code size inflates the headline coefficients. Per household, correlations fell by 0.10 to 0.20, so roughly a fifth to a third of the raw-total association reflects zip size.

Short and incomplete time window. Zip-level LIHEAP data cover only January to July 2023 because about 74% of later records lack zip codes. Seasonal and policy effects cannot be separated.

Observations are not independent. Repeated months per zip make confidence intervals and p-values too narrow.

Within-zip correlations do not remove countywide trends. Countywide monthly shocks may drive part of the lag pattern.

Measurement and aggregation. Food bank pounds are recorded where food is distributed, not where recipients live. Results describe zip codes, not households, and isolated significant results from many tests should be confirmed before action.

Conclusions and Next Steps

LIHEAP enrollment tracks food assistance consistently across lags, policy regimes, month exclusions and per-household rates, most strongly for CalFresh. Use it to identify zip codes likely to need assistance, especially high-poverty zips for food bank outreach, rather than to forecast changes. Next steps: report per-household rates, add month fixed effects and cluster-robust errors, and extend the analysis once later zip-level LIHEAP data are recovered.